ActInf Livestream #023.1 ~ "Embodied skillful performance: where the act

Livestream | 1:41:39

greetings
hello everyone thank you for joining
live or in replay
today it is may 27th is that true
not not even close not even close
it's june 8th
it's june 8th 2021 we're here in actin
flab
live stream 23.1 and we are discussing
the paper embodied skillful performance
where the action is
so welcome everyone to active lab we are
a participatory online lab that is
communicating
learning and practicing applied active
inference
you can find us at the links that are
here on this slide
this is a recorded and an archived live
stream
with all of the advantages and
constraints that that provides
please provide us with feedback so that
we can improve on our work
all backgrounds and perspectives are
welcome here
and we'll be trying to follow good
etiquette for live streams
from this short link you can access
the calendar of events for live streams
of multiple different kinds and
highlighted there with the blue arrows
are
today's discussion on june 8th and next
week's discussion that'll be on the 15th

and they're both going to be on this
paper embodied skillful
performance so we had a good time making
the dot zero and
learning about this and kind of opening
up what this was adjacent to
and now we'll look forward to continuing
the discussion and just seeing where it
goes
for those who are watching live it would
be i guess
especially appreciated to throw us some
questions whether they're
related to the paper or not
the goal for today is to discuss
and learn the paper is embodied skillful
performance
where the action is by hipolito baltieri
fristen and ramstead from 2021
and today in 23.1 we'll be
uh ambling skillfully we hope through
the paper again
and flagging areas where we want to be
talking with the authors next week
greetings dave
welcome and just seeing
where we get to opening up threads and
kind of
enjoying the middle part of the 1.012
sandwich so we can begin with
the introductions and the warm-ups
today um there's a few of us so we can
go around in the introduction and just
give a short introduction or a check-in
and we can also
then pass somebody who hasn't spoken
taking a look at the warm-up questions
so i'm daniel i'm a postdoctoral

researcher
in california and i'll pass to dean
yeah hi i'm dean um i'm up here in
calgary um
and i'm getting more and more involved
with this
uh idea of sort of building up the
active inference profile i'll pass it to
david dave is there
but if he isn't i'll pass it back to
daniel
dave you're welcome to join in whenever
you would like
i hear you tapping a little bit but
maybe whenever you want to
speak otherwise what is the active
inference profile
that's what i think we're trying to
build out here i think we're trying to
give that some anchors and some leverage
so
we haven't figured it out completely yet
it's like
everything in active inference it's it's
a generative model
it's growing and i don't know what its
final form will be
or if it takes on a final form even
are there other areas that have profiles
or are there other
profiles of note
yeah i i think there's convention
that that's already an established
profile and i don't know that this is
has got to that place yet where it where
we can say it has
a convention or something that's
predictable about it and i think that's

what makes makes it kind of
uh well in my mind it makes it curious
where is it going as opposed to where
it's been and what it what it's
established so far
nice points agreed active inference
especially
in these nascent times it's like a
strange attractor
where people come in from so many
perspectives
and then they get slingshot out in a
different
direction or with a different um memory
or different
view so dave no worries about the
hardware mystery
everyone is welcome to just type in the
youtube chat if they have questions
so i think for the rest of this stream
we're going to walk through again maybe
thinking again about walking as a
skillful action and
all the fun similarities and differences
between motor
actions which are what are discussed
here and that sort of space where motor
and knowledge come together like with
skilled action
yes dave your your uh different
embodiments are muted indeed
so the paper uh
laid out their aims and claims and the
central aim of the paper
according to the authors is to discuss
critically the limitations
of instructionist control theoretic
models of skillful

performance so the big dichotomy or
dilemma that's going to come up again
and again
is this difference between
interactionism and instructionism
so maybe dean where are you
coming at that from or how would you
describe the difference between
instructionism and
interactionism well
well first of all even including
interactionism as a formality
i think is a huge step it's
metaphorically speaking it's
it's opening both of your eyes you have
a second eye available but
maybe you weren't aware of it and now
with with the idea of turning that into
sort of a
a structured or formalized approach as
opposed to
just the instructionalism piece
which is traditionally what we've
formalized
is it's not just a doubling of
of what we might be aware of it's it's
actually
placing ourselves where we can see the
depth
of what we're observing or what we're
analyzing or what we're addressing
and i think that that's that would be a
huge step if we could get people
comfortable with that
i'm not saying that people will
necessarily get comfortable with it
though
because for a lot of people let's be

honest um just following along and being
instructed is difficult enough
and now you're introducing a whole
second lens
of information and very quickly you'll
have a bunch of people screaming
no you're going to overwhelm them you're
going well i'm not
just because i have both my eyes open
doesn't mean that i'm going to be
i hope it doesn't mean i'm going to be
scared i hope it means that i'm going to
feel a little
like a little bit more grounded and
centered but we'll
i don't know we'll we'll have a
conversation about that today especially
when we include the whole body
yes the entire corpus one thought there
instructionism
being like instructions are passed from
a to b you know water flowing downhill
instructionism has a directional
uh context and also the idea that what's
being passed is the sort of packet of
meaning like here's the envelope with
your instructions now carry them out
and what i think is exciting about
interactionism is it's not just
a bi-directional instructionism it's not
like well you instruct me and then i'll
instruct you
that's sort of the the low bar for
interactionism but where interactionism
really blows the lid off
is when we open into that third space
of interaction and improvisation so i
think that's what's kind of

interesting to draw out and they are
going to highlight in this paper how
control theory models of skillful
performance or otherwise
implicitly or explicitly do have
instructionist assumptions
specifically this idea that motor
representations
convey or harness instructions about how
to perform a specific
task so even if the authors of those
control theory papers didn't explicitly
use the word instructionism
then they're still playing into this
idea that instructions are being passed
for example from the brain to the body
in order to be
carried out and we'll be looking at that
from a more formal perspective in a few
minutes
and the big claim that they're going to
be reaching is that
active inference doesn't need to posit
that instructionist assumption
so there's a way to frame control and
skillful performance in the active
inference framework
that doesn't take on these
instructionist assumptions and the
content
based directional uh message passing
that it entails and
that'll be something fun to explore also
dave has a nice comment in chat
dave wrote as i recall piaget and
colleagues counted and described
something like 400 distinct embodied
concepts

used by the walking skills of a typical
15 month old
so that just shows how there's a lot
going on even with uh
behaviors like walking cool
it's one other thing too the assumption
that
somehow uh the the
muscles and the nerve endings sending
signals that are in symbolic form
is quite a stretch i mean we talked
about that
last week when we got to that inverse
directionality slide
but i thought about that a little bit
after because i went back and watched
after last week and i was just like i
know
for me i was incredulous a couple a
couple of times we went through the
paper with with blue but even going back
and looking at it
you'd have to you'd have to build a
really
powerful argument that says that
the pressure signals on my fingertips
are coming back to me
in verbalized form
exactly it's this idea that uh you know
the files are on the computer
like what's coming up from the
fingertips is pain and what's going down
is an instruction that's sort of the um
cartesian dualism mapped on to
instructionism
is just like body and brain are two
disjoint pieces and they pass special
symbolic information to each other right

and we'll see our brain and our bodies
having a conversation
so what is it if not a conversation or
could it be beyond a conversation
right right nice cool
we covered the abstract last time
all we'll point out here is just that
they start as
good philosophers with an examination of
the instructionism
and what that entails the second and
third sections of the paper
are characterizing the representationism
representationalism that's related to
these motor commands kind of what we
were just alluding to
and a special focus on the optimal motor
control
theory and then the final sections
take it to predictive coding and to
active inference which are
formal frameworks for modeling behavior
that are related to
control theory formalizations but
the goal of the paper is to show that
some of the instructionism
is left at home which allows us to
go out with interactionism
here is the roadmap framing a similar
uh outline and also
just giving a few more uh
signposts along the way with the figures
and the box and whatnot
the keywords provided here
were helping us understand what did the
authors
want to have their work indexable near
in the semantic net of literature online

how did the authors want their work to
be
grounded and it was in skillful
performance
that's in the title optimal motor
control that's the sort of
um scapegoat or foil that's the contrast
maybe even bringing us back to
contrasting
divergences instructionism again
tied up with the optimal motor control
motor
representations lending us into
action-oriented representations and
finally
taking us to active inference where we
kind of
begin and end
so skillful performance can mean many
pieces and the authors defined it
as opposed to bear movements such as
breathing and blinking
skillful performances are intelligent
bodily activities
which harness knowledge about how to
perform certain movements expertly
it reminds me of the eternal debate who
slash what
is intelligent if an insect knows how to
walk
or if it is trained to do something
versus if it's something that
its body provides the training for
inherently what is skillful so
definitely
will want to ask the authors and anybody
in the chat and anybody who joins us
next week what is skillful performance

what is unskillful performance if
somebody tries
to do a skillful performance and fails
is that still
within this category of behavior
i saved i saved up for this week because
i wanted to get to the
after the comma the witch harness which
harness
knowledge uh like i kind of
hinted that that's that i wanted to ask
a question and i still do want to ask a
question well
what does that mean to harness knowledge
um consciously or
or is the can you get to a place i mean
blue talked about this you talked about
this
can we get to a place of doing it
automatically
does that mean it's harnessed just
because we're not thinking about it
and and how much time did they spend
looking at that piece because
that that kind of demarcates out
what skill is is it if it's a dependent
if it's a dependency on harnessing then
what
how do we define that how do we define
what's what's captured and what isn't
and the harnessing brings up kind of
like harnessing or
yoking uh you know buffalo or a horse or
something like that
and that's so kind of hinging on that
word
it's like the knowledge is that wild
stallion and it's actually the bodily

activities which harness that knowledge
because when you quote know
that your body can move in many
different ways but skillful performance
is when that is harnessed it's when it's
connected to an instrument or whether
it's its own instrument
and it's channeled in a way
maybe that's productive or expressive
well at a fundamental level we have to
again be careful
that that something going on at our
fingertips is being
harnessed by something going on
three and a half feet away yes connected
but
but is that an assumption that's
verifiable
is there an actual harnessing going on
even
because lots and lots of times things
happen at my fingertips that i'm not
sending i'm not even aware of what my
hand is doing
my wife tells me stop jiggling
right just because i got nervous energy
or whatever
i think that's going to come back also
to the predictive processing
and predictive coding frameworks which
is sort of
when the body is getting the same the
signals it expects
like if you're not consciously bringing
your attention to
you know your foot if it's just your
foot in the shoe
it feels like nothing or there's no

attention paid to it
and you can kind of shine that light of
attention on it and it will feel
normal or maybe you can start to zoom in
on other sensations like tingling
or other feelings but the sort of
default state when you're not paying
attention
things that are happening the way that
you expect is
neutral and so that's what that's why
that active inference piece is really
interesting
because in order for it to be active
inferencing there has to be
some kind of attention or awareness
being paid
and that's like it's attention as um a
resource
gotta pay the cost of attention
or suffer suffer the regret of not
paying attention and dave
dave wrote trying and failing especially
if part of the problem
is choking which was brought up by the
authors in the paper as well
is likelier to involve more neocortical
activation
and more access consciousness than
smooth performance of the same
routine so trying and failing there's a
lot more
inner dialogue or thought process like
awareness happening whereas when
something is performed skillfully and
smoothly
it just sort of goes off without hitch
and

just you know you whip out the
instrument make your your shot or your
swing or whatever it happens to be
and then maybe you can narrate in
retrospect but
in the moment you're not having as much
awareness
so that's something one last voice oh
yes
so this is really interesting to me
because again
um is it skillful to be in the peloton
and be
be pulled along by what we would
describe as
the intention of those in front of you
is that considered to be a part of this
because there's a lot of things that
we're aware
and conscious of and some things that
happen in these skilled performances
they're just because they're part of the
collective they're
they're caught up in the inertia of the
moment
and is how do we how do we
factor that in when we're talking about
active inference because i think
that's what instructionalism
[Music]
doesn't attend to there's many many
things that are going to get dragged
along here
just because they're drafting
and how do we how do we address that
when we're talking about
skill performance skillful performance
and interactionism because

interactionism at least says i can be
caught up in an
in an inertia moment i don't have to
explain it
absolutely um instructionism
has to pitch and hold everything into
that symbolic
information theory what is the
instruction from the front of the pack
of the flock of birds or the peloton of
cyclists what is the instruction that's
getting passed back
and so we have to go from what we know
is just happening which is like the
physical
inertia there's also a narrative inertia
there might even be just an air current
going over the person that actually
stabilizes them
in their um spot we don't need to fall
back
to referring to those as instructions
when they're clearly interacting
phenomena
different types of interactions you
could have uh somebody breathing heavily
next to you and that could give you a
signal but it's about the unpacking
of that interaction and then where that
takes the collective rather than
again just thinking about everything as
connected with wires and
passing secrets right cool
um we can look again at the active
infrared plant so this was just
asking where are the limits of skillful
performance
if the niche and the fitness to the

niche is what counts for different
organisms
our human cultural niche has all these
awesome
kinds of skillful performance and
otherwise but what about
other creatures what about other
phenomena the kinds of things that we
might
want to also model with active inference
but just don't
quite have that same level of um
maybe thinking through other minds as we
have with people whereas we try to put
ourselves in the shoes of somebody else
do you think daniel that that plant has
um
got an embodied technical
skill
welcome blue hey blue hi
we're just wondering whether plants have
the um
this plant okay but i have a technical
skill now that it has the
the tools to be able to move from side
to side
it's a great question
um
i wonder if if it could be shown that
it got better at moving from side to
side
that would start to look increasingly
skill like
and i think it's it's that idea that
it could uh the molecular changes could
occur that would
allow the plant to respond faster or you
could imagine selecting

on a whole greenhouse of plants and the
ones that are able to switch
faster those are going to grow faster
and then you select the ones that grow
faster based upon their ability to learn
in that generation
so it kind of shows how you get this
development of skill
within a life that's development
but then between generations you can
also have
the um entrenching or the canalizing
of learned skill like language
and skillful performance in in this
paper and in our
slides it's kind of like we're hinting a
lot at this full body
type performance uh high jump and things
like that
but of course language
is maybe one of the examples of skilled
performance it's a motor
behavior it involves a huge number of
interacting
motor subunits voice box and the mouth
and speech therapists and others
they go deep into that but
speech is almost like the case where you
say well surely
instructions are being passed that's
maybe the
the high water mark for where
interactionism could go
would be to have a speech a framing of
speech
that moved us beyond instructionism
because this seems like clearly the
transmutation of

semantic or symbolic information into
not directly related motor behavior
so that would be kind of a nice case to
investigate
and dave wrote in the chat it seems
odd that dogmas that have been known to
be false for decades are still premises
of so much funded professional
activity i'm thinking of bf skinner's
insistence that all control activity
must be strictly cortical
even though it's been known since at
least the 50s purely on the basis of
speed of nerve conduction that that
would prevent graceful walking
let alone faster and more intricate
movements so that's pretty
interesting we have
almost just anatomical reasons
to know that not everything can go
through this sort of control
and cognition in the loop there's just
not
fast enough transduction along the
nerves so there has to be this
sort of distributed function
more consistent with an interactionist
framework than an instructionist
framework
yet it's easy to fall back to
instructionism maybe especially for
professionalized areas
let's go to um optimal
control blue if you wanna you know feel
free to add anything that you'd like
optimal control theory is a nice bridge
point because in the paper and
as we'll be going into we're again

thinking about motor behavior
but control theory is something that
applies to
the electrical grid to logistics
to robotics which are i guess
motorized but optimal control theory
is a framework that expands to many
non-motor systems as well
so that makes me wonder about whether
we can also apply that instructionism
versus interactionism
contrast to control theoretic
situations that
are implicitly instructionist like
there's an algorithm and then it sends
the buy
order to the unit that makes the
purchase
on a market or it sends a certain
piece of information to a different
computer that then enacts
that like a suggestion
and then even in systems where we know
instructions are being
sent like computer systems
maybe there's an interactionist way to
think about that
too any other
thoughts or does optimal control
essentially place all of our
chips on the top down
[Music]
uh aspect of of the relationship between
top down and bottom up
does it does it do it does it optimal
control
say optimization happens exclusively
from

a top-down directionality and that there
is really no
place for the bottom-up part
i think it's a good question and i'll
skip to this
fristen 2011 paper
which is entitled what is optimal about
motor control
right i wonder if um
that optimality does arise from the top
or um
what is optimal once we move beyond
instructionism there's a perfect set of
instructions there's a perfect way to do
it
and then that is either carried out
faithfully or it's not
that so philosophically if you said
optimal would be some some
um openness to both
an ability to recapitulate and an
ability to discover
to be able to put those two things into
some sort of
coherent relationship um
i'm not saying that that's what is being
said here but
well i'm just not sure i'm not sure why
we always assume that optimal
is always about being able to create
copies
even in scope of skillful skillful
performance
even compared to a standard there are
things that happen when a gymnast
does their routine where some
some element of the the
variability that applied to that

situation
is considered when something is being
judged i think the only thing that
that doesn't doesn't factor in
variability is the clock
right like you either ran 9.8 seconds or
you didn't
but many of the skill skills skillful
performances that we
pay attention to aren't always about
a recapitulation so
it's interesting i don't know i i guess
again it's how you oh how you define it
but we want to we actually want to get
on to doing something useful here so
yep and also here's a line in that first
in 2011 that reminds me of
dave's comment which is uh these
estimates which are coming from a
predictive processing or active
inference framework can finesse problems
incurred by sensory delays
in the exchange of signals between the
central and peripheral
nervous system so the generative model
can include
space for that delay when we think about
just sort of
two people who are talking and
accounting for each other's delay for
example in that kind of an interaction
versus the send process
weight send back model
there's actions that happen faster
another example that maybe would be like
baseball
where there's such a small amount of
time to decide whether to initiate

the behavior of swinging
and this paper from 10
years ago laid out just such an
interesting
difference between value learning
and active inference which allows for
that
pragmatic and the epistemic component
the optimal suggests multiple things
first off it's like it kind of suggests
that there is an optimum that is being
achieved so that's one maybe easier
way to one easier point just say okay
well maybe there's an optimal way maybe
there's not an optimal
way but what is being optimized is
the value just like reinforcement
learning
what's being implicitly optimized here
is
a value function and just as it says
here the replacement of
value and cost functions with prior
beliefs about movements
removes the optimal control problem
completely so this idea of the value and
cost it's kind of like an economic
analysis
of motor behavior so then let's just
imagine that any
number of circumstances like somebody um
has a
belief about how a performance will be
viewed
socially well you could couch that in
reward and you could try to
go break down from the narrative and the
social all the way down to the movement

of the fingers on the piano
about the value and well if you mess up
it's a lower value action
but you're getting really far away from
the embodiment of the performance which
maybe doesn't have that same sort of
cost all the way down
so it's like will we have cost functions
and economics all the way down
and value and cost or will we have
generative models and an instrumental
way of thinking about
action blue anything overall
otherwise also anyone who's watching
lives welcome to ask a related or an
unrelated question
i'm good anything after last
week's discussion that you'd been
thinking about
um i'm trying to remember last week's
discussion
refresh my memory yep that's what this
first part is
let's go to figure one and two of the
paper
and take a more
formal look at uh and we can go to first
in 2011 if we want
more details on the similarities and
differences because this is kind of the
core
of the contrast between opt
optimal motor control theory and active
inference it's going to come down to the
differences
in how these models are framed because
there's an interpretive
layer but at the heart of it it's the

differences between
these two kinds of models between figure
one and figure two
here so in figure one
uh and we'll go to the juxtaposition
in figure one and figure two we have in
both cases
the plant kinetics which are defined
with the exact same
equations and a sensory
mapping so this red box
is the same between both of the models
between control optimal motor control on
the left
and active on the right and
there is still an optimal control
module in active inference so that is
interesting but if we look a little
closer
we can see that in figure one
that uh motor commands that u with a
tilde over it
it's the minimization so it's
optimization
minimization maximization sometimes by
minimizing the cost
you maximize the profit or it's
flippable throw a negative sign in there
and a min becomes a max
so here what's being minimized
is an integral of
the potentially the cost function c of x
and u
for changes in time so cost is being
minimized through time and that is
optimal control in that framework when
we look at the optimal
control in the active inference

model we see a really different
structure
even of that optimal control module
we see that u which is playing a similar
role
it's not a command but it's still being
passed this
u of t function is still related to
a minimization but what's happening
inside of the equation
is very different and we can um
go to the first in figure caption to
see what is happening there that
e curly e is the distinction between
exteroceptive which is e with little e
and proprioceptive e with a p reporting
on hidden states
in intrinsic extrinsic and intrinsic
frames of references respectively
these prediction errors are the
difference between sensory input
observed and
predicted so going back to here
what's being minimized isn't the cost
function
through time even
philosophical disagreements with homo
economicus aside okay
let's put aside cost and then then
there's the challenge of assigning costs
to different kinds of states
what's being minimized here is the cost
through time i mean don't businesses try
to do that everyone wants to figure out
how to minimize
cost or time it's challenging to do
what's being done here
the minimization between the

proprioceptive
sensory differential with a
multiplication potentially
through time not sure what that does in
the model
but it's just so interesting that um the
minimization
here is just like we've talked about in
so many of these discussions it's about
your generative model
of action and sensory outcomes sensory
outcomes are emitted by hidden
underlying states that you can't
directly observe like you don't know
exactly whether it's night or day but
you are getting photons or not
so there's a hidden state that's being
evaluated
that's the sort of hidden markov model
component and the hidden state is
in the model emitting states that are
being observed as sensory outcomes
and in active inference the depth of
that generative model
is actually such that instead of having
a really rich generative model of cost
functions
and then trying to reverse engineer your
cost function to find the cheapest
behavior
the generative model is of expected
sensory outcomes like every time i move
my shoulder here it starts to hurt
and we don't need to introduce a concept
of
reward or cost into that equation
we can suffice to say that there is a
generative model of the sensory outcomes

under different motor policies and
the organism has a preference for let's
just say being
in non-painful states
it sidesteps that question of cost
in a way where we can see that even from
the side by side
even where there is an optimal control
element like a
minimization element what's being
minimized
is something that is very very different
and that's one of the key aspects of
active inference that
what's being minimized is the divergence
between
expected outcomes and realized outcomes
that's what of course makes it so
related to predictive coding predictive
processing
rather than drawing only on the
reinforcement learning
type approaches that are doing a
minimization
implicitly maximization on reward
daniel can i color in a little bit too
please when i saw this it was
interesting because
what i saw was a rep not
i don't i guess it was a replacement so
on the figure one it's
do estimation
and then in figure two what i saw was
the
removal of that and the
application of a rule so
uh minimize
uh prediction errors that's the rule as

opposed to a step
and all i thought about was for example
hippocratic oath i i don't know what i
should be doing in terms of this patient
but the rule is do no harm
so i think that's what i thought was
really interesting here is that
an actual step was replaced with a rule
and that and that to me was the big
differentiator
the introduction of rules that's why i
asked last week
when you changed the rules of the game
for the plant
i know the affordance has changed and i
know the tools were introduced and all
of that stuff
but i i keep coming back to tools rules
and pools
tools being obvious rules being obvious
pools just being an aggregation of
resources
and so i think if you if you're able to
sort of
coordinate all three it gives you a
different sense of what might be going
on
okay a fun thought on that just to
think about game playing the rules are
always in effect
and the rules can be explicit or
implicit like it's a rule in chess that
you have to move out of check or figure
out how to get out of check
but then there's sort of what are
equivalent to rules like if a piece is
pinned it can't move
but you don't need to state that as a

rule so rules are they're simultaneous
they're always in effect unless there's
another rule and
they can breed emergence you just say
okay there's here's how the pieces move
and then here's check
those are the rules and then it's as if
there's a rule
that is you can't move out of a pinned
piece
so rules always in effect allow for the
space of emergence
steps have to be sequential and so it's
almost like we're moving from
the um within a round of a game
that's the instructionist okay you're
playing risk first
roll the die now see whether the three
is higher than the two die or whatever
it happens to be
and interactionism is pulling us back
to like the rules in the pattern that
still
there can be sequential instructions or
sequential information can occur
but it is very different to have
rules rather than specified steps so
let's see
we're building up yeah go ahead i just
want one last thing so when i would ask
young people
to watch a video clip of something and i
would ask them to
draw three columns and title them tools
rules and pools
and find examples find tokens of each
what do you think was the easiest thing
for them to accumulate under

it was tools yes those those have got
the edges
what was that what was the second
highest aggregation
pools because they could they could
literally figure out
what was congregating what was the
hardest thing
the parallel processing which is what
you just described
it's always going off the sequential is
much easier
to account for the parallel processing
the happening as co-occurrence
is the hardest thing for people when
when you're walking into a novel
situation
to take account of so that's you can
you can say it it's just being aware of
it and finding examples of it
that's really really hard for most minds
so even from a computational aspect
right like the
parallelizing functions make them like
speed up so much right but
it's they also like are so
computationally intensive right
yeah exactly dave has another
great comment on this slide he wrote um
i'm looking for where time is applied in
the two approaches
the intrinsic frame of reference for
active is
u tilde of t so that's um we'll
zoom in a little bit on just figure two
here's that u-tilde of t
passing through the intrinsic frame of
reference um

but there's no explicit t in the optimal
control formula we can just see that
motor commands utilda are passed
so there's kind of like a timelessness
to con you know contract this muscle
it's not contract this muscle now it's
actually just
um a timeless statement so
t he says whereas t is inside the
optimal control
formula um here is t
uh dt it's a derivative through time
and not in the intrinsic frame of
reference
as seen in active so he asks whether
that tells us where
the approaches are taking account of
time differently i think that's a great
question is how does time play um
a role differently in these two
frameworks let's keep on
looking for similarities and differences
so we have rules not steps for active
inference
we introduce a time dependence of that
downward
motor signal um we have of course a deep
generative model component
we have state estimation is that's the
um the hamlet joke there's this whole
module
that's clearly different in optimal
motor control
which is the state estimation and that's
what doesn't need to be done
in active inference is this state
estimation
of okay given the sensory input we're

getting
like i'm getting um you know two-thirds
of my proprio receptors here and
one-third here are firing so
let's estimate that the arm is here okay
now
pass that to the next part of the model
given that the arm is here
what do we want to get the arm to
okay now given which side of where we
want the arm to go to
we estimate the optimal and the actual r
send the command you know turn the
steering wheel to the side
to bring it into alignment with the
optimal or the most rewarding state
so state estimation is in optimal motor
control theory
but an active inference you just don't
have it
and that's sort of an interesting piece
to
drop out of the model because
we don't need to have an explicit
representation
or an explicit variable with the
location of the arm
so that's a rule
it's like john cage everybody has the
best seat
it's almost like it's enough to just say
you're on the ocean you're sailing and
then
you're going to be trying to reduce your
uncertainty
as you pursue policies that in your
generative model of the world which
includes latent

causes so the degenerative model
includes like latent causes of the world
which are not anywhere to be seen in
this
um optimal motor control theory like
there's a deep latent cause that if too
much weight is on
you know this branch it's going to break
but
where is that the constraints of the
niche
or something that's deep detonation not
there so anyways
it's like you're in the boat you want to
be reducing your uncertainty
about getting the sensory states that
you want like
seeing that land look bigger reflecting
you being closer
but you don't need to have that gps
coordinate
where am i okay now where do i want to
be and how are we going to get there
it's enough to actually have a sensory
preference
and then reduce your uncertainty on the
trajectory of reaching
that sensory outcome so that's
a big difference in active inference
and i'm seeing even a few more
let's look at the green boxes so in the
green box
for figure one we have a forward model
so they write that the function of the
forward model
is to improve the execution of action
it's kind of funny executive
executive office or the executive

component of the government it's about
executing that's where the instructions
are issued from executive orders
improve the execution of action by
helping to finesse the inferences of the
state
estimator so again it's in feedback with
the state estimate and then it's sending
an effort copy
so that's outgoing is efference
and so there's this feedback between the
prediction
of um basically where
the body is in this case and then
that gets sent it's kind of interesting
that the x comes from the state
estimation that's the where
is the arm and then the forward model is
the forward model of motor control
that is what connects with this u tilde
that's the green box that forward model
in
optimal motor control whereas
in active inference we have a different
forward model it's a forward model of x
as well as v which is the prior belief
so again there doesn't need to be
um a forward model of position and motor
action
as happening here it's enough
potentially to have a model of
action um and over beliefs
we can look at the first in um caption
to see
a little bit more closely
uh yeah any thoughts on that
i think the state estimator to go back
i think one live stream state estimator

says that there have to be ties between
the two rails
to stabilize those ties so that some
processing can take place
i think what this talks about is the
fact that we know that there's a
gradient
we know the train is going to roll
downhill
as long as there's a gradient it's a
rule
as opposed to trying to stabilize the
relationship between the ties
of the railroad two very different
ways of looking at something and i think
that
um this goes kind of back to what i said
i think there are times
when we do give ourselves instructions
and then there are times when we simply
parallel
process because we have a rule i don't
want to make mistakes
because that doesn't serve my my goals
or my target
and i think all i i again i just suggest
that the inclusion of both
is probably to our advantage it's just
that what we tend to do
is give all of our we've been trained to
give all of our
attention over to making sure that the
thing is stabilized
nice also one piece of first in 2011
that's not
brought over to this paper figure one
was that motor control
figure two is predictive coding and

motor control
so already we see the reduction of the
model
in the terms of the state estimate goes
away you don't need to explicitly have a
state estimator because the state
estimator
has now been subsumed into the forward
model which is why crucially
the mapping from hidden states to
sensations is now part of the forward
generative model
so we have a generative model of what we
expect
and now with predictive coding instead
of keeping a variable of
what we think is happening we just have
which is can be tracked through time and
all that
we're simplifying that state estimator
by including
the expectations about the state
estimator in our forward model
that's predictive coding how does that
differentiate
from active inference so that's a good
question
here we have plant kinetics sensory
mappings
forward model and optimal control with a
cost function still coming in
you still see that integral with the
minimization
of motor commands u over
time whereas in active inference again
we're moving from it's more valuable to
be getting the signals i like and i like
rewarding things

to i'm just bringing the sensory
outcomes
into alignment with expectations and i'm
selecting policies based upon which ones
i think are going to get me there
and then this section that um
i had highlighted in summary the tenet
of optimal control
lies in the reduction of optimal motion
to flow on a value function
like the downhill flow of water so
that's pragmatic value
and that's awesome because think about
how much work
has been done on information flows in
the last 10 years
conversely in active inference the flow
is specified directly in terms of the
equations of motion that constitute
prior beliefs like patterns of wind flow
the essential difference so how do we go
from
water going downhill on the value you
know gradient descent
or up and down again they kind of get
flipped around but
how do we go from just thinking about
what direction does gravity take you
that's value that's reward there's one
answer you know which policy
for financial outcomes is going to have
a higher
value that's that scalar estimate of
value
the essential difference is that prior
beliefs in active inference
can include solenoidal flow so this is
where we're opening up the whole

discussion of play so i hope that we can
think about that for a second
and fristen says you know he doesn't
want to overstate the shortcomings of
optimal control
so it's not that you can't introduce a
solenoidal component
to value you can have the idea of
solenoidal flow
circulating on iso contours but how many
businesses stop
and say you know let's not pick the most
rewarding number
let's look for maybe it's not the most
rewarding number but then there's the
most
solenoidal play around that number
people say but we're not going to be
doing those so
why would we stop at you know 5 000 feet
when we know that we can get to 6 000
feet and
solenoidal flow and the ability to have
that
pragmatic and epidemic that directed and
the solenoidal components
opens up the space of play hopefully so
what do you think about that or does
that
i'll put up this slide where we had
separated out this optima motor control
theory with a value only
that's the one thing that this robot is
tracking is
the slope and the altitude it's
following is uphill as much as it can so
it can get to the top of
mount value whereas in active inference

we're using the helmholtz decomposition
which is related to the physics of flow
on vector fields that is going to reduce
to these two irreducibly
separated components so like they're
totally orthogonal they don't have
redundant information
they have totally separate information
there is the value component the
straight line
but there's also this equal value
solenoidal flow
any thoughts on that
you explained it really well here's a
comment from dave if while
people are thinking um giving oneself
instructions
can refer to something real and useful
if one interprets the execution of the
instruction appropriately
i'm referring to playing a piece of
music without touching the instruments
which can be nearly as effective as
actually playing
if the challenge is to simply touch the
right stops at the right moments this
can actually be
more affecting empowered machine
operators may seem less efficient
than highly regimented ones but may over
time get to far
more effective overall operations
dean what do you think about that again
i completely agree okay
yeah so my my only question is so why
are
more people not picking up on the
potential of this and the value of

having both
an awareness of both
so i think like it's maybe just getting
started in a computational
way i mean we've we've explored some
computation
papers um looking at you know
how to optimize a given task
computationally
so i think that it will um
gain traction right it's just so new
i mean not active inference as a concept
but um
the practical applications of active
inference i think are um
just starting to be explored and i think
that the play
um aspect is is
important i mean because it's it's like
how we learn right exploring or playing
or or
um you know these types of things give
rise to to real learning
uh even if it's accidental learning like
not
necessarily effort directed
yeah i think i think it's also
i think it's something some has to do
with um
a perception that
play doesn't necessarily generate value
which is kind of sad because there's so
many examples where it does
the outcomes of play caused your mind to
go off in so many because it's supposed
to be about distribution
this is a kind of a way of of of
encouraging that distribution and yet

for some reason it just doesn't get the
same
respect so
i'm just thinking about the um pipeline
from basic to translational to applied
to sort of infused research
like once oh everybody knows that you
know that what
what was once basic though becomes
applied and then becomes
every day so let's just imagine that
we're going to shooting on that skilled
performance slide
well if we have implicitly or explicitly
if we're working under this
optimal motor control function we're
thinking
okay maybe people's biomechanics differ
but for each person's biomechanics there
is an optimal way
to shoot a free throw and we want people
to have a more accurate
modeling of where their hand is on the
ball so that they can execute
the movement better so that we can get
the reward you know score the points put
the ball in the hoop
and so it relates to
having people practice or maybe get
input on
your hand was a little bit off here
relative to the coach's
expectations of where the hand should be
um
but then there's this whole question
about visualization
and about what dave was saying like
playing the music without touching the

instrument
and the ability that was almost
discovered in a bottom-up way
of changing beliefs about one's
performance
rehearsing mentally but also changing
beliefs about performance
influencing motor behavior that
is more consistent with what we see in
active inference
and so there's just multiple things that
previously would be seen as sort of
exceptions
or quirks of a pure motor control
system like incorporating these
deeper elements of our generative model
um
all the way up to even like the
psychological like maybe your generative
model
is that your team is fated to lose so
then you do choke in that moment because
like
you know you didn't psychologically want
to win or whatever
and that's as gina's we're always joking
about like the interview with the
athletes right after the game
it's like i wanted to win or something
like that like
they're not going to give you that third
person
my hand you know was on the wrong spot
on the ball
um it just we're seeing how
now as we have just like blue said more
computational modeling
that recognizes interactionism and that

recognizes deep generative models
it will be possible to start to slide
some of these examples from being like
quirky outliers
to actually sketching out the structure
of inference and action instead of just
being seen as sort of like
this one weird trick that's going to
help you learn free throws
it'll be like that's always about your
generative model
and that includes you as being a team
player for example it always is going to
include that
and then it's going to come down to
maybe
being modeled better by active inference
rather than a reward based framework
blue
so i just was thinking like you know in
your you said
you were exploring like the the dynamic
you were you were going through the
dynamic there was like exploration and
then doing something intentionally like
learning how to do it and then like
perfecting a skill
right i was thinking like you left off
the end of that like
uh just like from a societal like
standpoint right like we explore new
things like not thinking as an organism
but thinking as
as a social structure we explore new
things we set out the intention to
complete those new things we perfect
those new things
and then we forget them right like so so

does that come into play
in in this i wonder because i think
about scaling active inference a lot
like from
not in the way that we talked about
scaling active inference um in the paper
scaling active inference but but just as
as from uh like you know small
components of a system to systems
and you know in that scaling like like
so
are we missing something about about
forgetting right like like handwriting
if you think about like learning cursive
you know that was something like people
start out scribbling on paper little
kids you know they're playing and
they're drawing and whatever and then
they
they intend to learn cursive they
perfect cursive but but like as a
society we're not teaching that anymore
so they don't know cursive
and it's like typing like now we have
voice translations so we don't know
typing anymore i mean at these types of
things like are influential in a society
so
does that come into play here like where
i've talked about before
you know thinking about doing something
like entering the flow state right when
we talked about the flow last
live stream you know it's like you have
the intention to do something and like
you think about doing something but when
you're really a skilled performer
you enter that flow state and you forget

about doing it you just
do it it's just magically accomplished
so is that like
flow state comparable to this like
forgetting
of as a society right so so when you
enter that flow state and like you're
doing something magically like just with
your kinesthetic memory
and you're perfect at it does that
compare to society like forgetting a
task because we've already mastered it
like we're
we forgot about that we're on to the
next thing i wonder if that's relevant
anyway sorry if i rambled
no it's it's really nice here's what it
made me think about was
instructions again implicitly or
explicitly
they have to be remembered or forgotten
because if they're not remembered
along the chain or at least in the ram
of the computer it's like
they're never carried out if the
instruct if you started a sentence and
they forgot what the first part was by
the end of the step
they can't carry out the instructions so
instructions we think about symbolic
representation remembering and
forgetting but with interactions
it doesn't have to be remembered or
forgotten it's actually just experienced
like in the ant colony the interactions
between ants they don't have to be
remembered it updates the generative
model of each nest mate

and then they continue on their path
but it's actually just experienced and
that's enough
in an interactionist framework rather
than being remembered
i'm not sure if that gets at it i think
it's a great point about how
as something becomes pervasive and other
skills
and scaffolds are layered over it there
is a forgetting that can happen
but also just it's like we don't even
need to have
memorization of everything like i
remember
hearing from different college majors
and somebody would say somebody would
say what major they were
and somebody else would say oh well that
one always was too much memorization
like there was that was too much
memorizing and then for that person
it wouldn't be it was their major it
wouldn't be too much memorizing
it just seemed like it was parts of a
bigger picture and so they didn't feel
like they were memorizing little
factoids
but rather just interacting with the
material
not sure dean what do you think about
that well
blue you didn't say this but you touched
on something that's
mattered to me for a really long time
and that is
should we scale bottom up
or does it sort of exist in its own

[Music]
realm because that's the thing that you
basically said
i i i taught for a lot of time
and it bothered me greatly that all the
learning from
from haptic movement was disappearing
into a keyboard
so so so we topped down that one
and then we forgot about it because top
down
basically erased the bottom up we we
seem to think
that if we talk down it if we if we give
it a nap
we can now forget about it and i find
that i find that crazy because i've had
many quite
quite heated debates with people who
want
to take another bottom-up exercise where
we can really learn
something and get some joy out of
discovery
and say well that's i'm going to scale
that and i say leave it alone
leave it alone it's a bottom-up thing
what are you doing
i know you can top down it but why can't
we just
let it explore a little bit and figure
things out for itself
so dean you really touched on something
there with like the haptic
movement um and you know i
like personally learn super well by
writing things down like i take notes i
throw the notes away i never look back

at them again
it's the act of physically writing
something down that helps me
uh learn um and i mean sometimes i do
look back at my notes but but
mostly not i just take them to to commit
things to memory
and um you know my daughter was really
struggling with
uh learning her times tables right she's
like doing these flash cards and the
quizlet app and all the stuff
and i had her manually copy you know
like the times tables and
you know instantly like within a month
she had them down
um and just like destroyed the
standardized tests like where they do
math computation which is just like
rote memorization of your math facts and
how fast can you do it
she got like 100 and she's you know at
like
off the charts for her grade level in
testing so
um anyway it's it's really very um
real and actual but but in that
kinesthetic memory and this goes
back to that like motor command you know
and like the disappearance of the motor
command
and so this is like what i think about
like these two juxtapositions and i
think like
there is a space like where you do issue
a motor command so
like at least for me right like there is
a space where it's like okay

i need to put my right foot in front of
my left foot and you know walk along to
learn to balance like i'm issuing a
command
to myself to execute some task but but
there are many times like where you do
not
issue that motor command you i mean like
think about like if you've ever tried to
learn to play the piano like placing
your fingers on
the keys and stuff and then it just you
just do it right like there's no
command at that space and so so there's
like a mental a kinesthetic like
something that happens
like where you transition between having
to issue the command and
not having to issue the command and this
is something that i you know i'm excited
to talk to the authors about if they
are able to join us on a live stream
because i because i don't know where is
that step where does that step
get erased in active inference i mean we
aren't always
just we aren't always commandless i
think i don't know
so i'll just put one more one last thing
on that if
if i think of myself if i think of
myself separately from the neck down
everything from the neck down is giving
me the bottom up
my brain is way up here there's things
that are supporting that that are giving
me that chance to bottom up i don't want
those things

to disappear i don't want to be a head
on a on a cart
i don't want to be like that plant with
just my head
going back and forth between lights
right like when do we
stop saying that everything has to and
like i said
you didn't say this but i've had so many
people say oh
dean if you if you all the stuff that
you're doing is great and you
you create all these unicorns and all
these young people are able to go out
and contribute in professional settings
but it's not scalable and i go
what what do you mean it's not scalable
every one of these things is a bottom-up
example that it it can happen as long as
you don't try to
top down it to death
interesting the part about the body and
the
neck reminded me of ken wilbur's centaur
stage of development which is sort of
like
uh along the integral psychological
trajectory
there's a step where there it's like
half human
half horse and so it's like one step
beyond kind of
a human in dialogue with a horse as two
separate entities
there starts to be like a fusion in
his framework moving towards non-dual
framings
but there's that stage where it's like

uh the top down the bottom up but you
know our brain
is further out just gravitationally
so it makes it seem like it's the top
down
piece but you know what about a person
upside down
then is it bottom-up signaling well
you've already pointed em out he's on
the edge of a building back there about
ten slides
or people on the other side of the world
they're upside down
right uh and i think uh asking about
speech and sign language
could be interesting because i think
those are gonna be the challenge
cases for symbolic
the last bastion of instructionism
and symbolic messages being passed
maybe is happening through language
whereas with this sort of elegant
skillful performance model maybe you go
okay i can see where predictive
processing is coming into play
where it's not just about value okay so
if predictive processing is an
improvement
from reinforcement learning um or
from optimal motor control then i guess
i could see why active inference builds
upon
predictive processing so that's that's
the pipeline the active
down the rabbit hole pipeline for some
but i think
when it actually is semantic or symbolic
what is being performed that'll be

where on one hand there's going to be
resistance
because it's going to seem the most
instructionist but on the other hand i
think we're going to tap
into the richness of active inference to
have a deep generative model
including semantic information whereas
the semantics are not
in this motor control framing
so it'll be like double or nothing with
active inference
for some of the symbolic cases uh blue
yeah i just wanted to go back to what um
dean was saying earlier about
bottom up and um like i don't know uh
like using active inference and modeling
active inference in
like a bottom-up and nested way is super
super interesting to me
um and and i just i'm i'm wondering like
when are we gonna crack that walnut
because it's gonna happen i feel like
we're we're teetering you know
on the edge of that um what will that
look like
i mean yeah what will it look like or
what will be a system where we can
explore it in
i mean maybe the active in france i
think is maybe maybe daniel will grace
us with we get to do his paper one day
we can do it soon but yeah the only
thing i'll add is that we have to
people that are participating in this
and really are going oh my goodness
we're really on the cusp of something
just have to be willing to go over the

edge of the cliff
and i'm not and i'm not joking it's the
letting go of all the habits that we
formed to get to this place because most
people didn't get to we didn't
the three of us didn't get here through
active inference that came after a whole
bunch of things that were more
instructionalist and to what to step
back it would be much easier than to go
over if we've got the prospecting the
proposition
and the provisions in place we'll go
over the edge
and it'll all be fine it'll be actually
quite thrilling
i i really like that like uh sort of
that come to the edge
there's a different thought chain when
you're climbing the ladder to the high
jump
like we've all jumped off of something
that was
scary or something like that there's a
different thought process when you're
climbing
maybe because your motor behavior is
grounded you're on the ladder or you're
on the mountain going up
that's one mode that's actually like
approach
but then right at the end the mind goes
blank
and you know the mind will say jump jump
jump
but there's that resistance and that
reminds me one of the few times that
i've used

vr goggles um
it was like you walked into an elevator
this was like a scenario walk into an
elevator
and then it carried you up a building
and you could see yourself getting
higher and higher up
and then the elevator doors opened and
you looked down
and it was just like dropping off a
building
and it felt like my foot was just glued
to the ground
like i felt like i could not take a step
forward
even though i could inch my foot forward
and i could see that it was
that i hadn't moved that i was still
just on flat ground
it was like it just
stopped motor behavior to know that it
would be
resulting in some unpleasant situations
in the usual case
so dean what do we say to those
who are um you know not sure if they
want to
go to the cliff what do we say to those
who are walking towards the cliff and
then what do we do with those who are on
the edge
this is this is fantastic question
because this is
this is the nut of the of the problem
so so do you want do you want
instructionalism
extended into the active inference realm
or do you realize that the whatever

identity you take on in terms of
learning
with both eyes open with some
instructionalism and some interactionism
i i don't know what that will be for you
or blue or
or me i just know it won't be an
extension
of instructionism alone i don't that's
the counter factual here
i don't know what that will feel like
as different any more than you knew what
the difference was going to be in terms
of
releasing to gravity as opposed to
pushing against it as you were climbing
up those stairs i just know that if you
considered it to be
just two more stairs as an extension of
what you were already doing
you're gonna be disappointed so i think
it's that
that is the that's the big question here
as long as you don't
as long as when i used to say to kids as
long as you don't think that a field
researcher
is simply an intern outside of school
i don't know what that's going to feel
like and i certainly don't know what
that's going to look like
because i'm not out there with you and
your sponsor
so how can you let go of what you know
in order to realize this thing that
nobody can give you
a direct answer to that's that's a
that's a beautiful question daniel

because that
that for active inference is going to be
the
that's the rule that's going to have to
be discovered that that even people who
are spending a lot of time
focusing on this right now giving it
formalisms and and developing the
the way to formalize the math are going
to have to have themselves otherwise
it's just going to be an extension of
instructionalism and that would be sad
yep it's like jumping off it's not a
staircase in reverse
right and um i think
the idea would be maybe we already have
flight suit on
where it's like if you climb
you climb up with the provisions and
then you know when you jump
you're gonna be able to do something
that was just
qualitatively different it's gonna feel
different it's going to be different
it's going to be a different
action and
wasn't the point of listening to
instructions to change
to update to be better or to be
different
and then it's like by being different
and following that path
we become who we are like somebody said
i always wanted to be
skilled in this martial art so then they
trained along the way to become who they
wanted to be and
that was them being who they wanted to

be
it's a lot of sort of loose
associations but again i think we're
working towards making it happen
it is it isn't though because
instructionalism my point was that
the instructions stopped i cannot give
you a description
of what that will look and feel like so
you're going to have to interact with it
that's there's not it's not a negotiable
if you really want
to be there you have to let go of the
instructionalism
so i don't think that's a loose
association that's a non-negotiable
now you may not like it and you may not
go there you may step back from the edge
and that's fine but don't think that i'm
or other people that are actually
looking at active inferences as a second
eye open
won't be aware of what's
what you're transfiring whether you're
going over
or whether you're stepping back that's
not a loose association that's
that's pretty clear
blue any thoughts on that i
i guess my question would be let's just
stay with this sort of like
climbing the instructions to get to
the space of interactions
what yeah blue god so it just
as like climbing the instructions to get
to the space of interactions like i feel
like
something is there that is has to do

with like the bottom up and the scaling
like you know in the systems like we are
climbing
the instructions as you know parts of a
system until we get to
to that space as a system
the rules well it's like okay we have
the ladder
going up or you know staircase going up
to the jumping off point
you could frame that in terms of
instructions
go to the second step go to the third
step go to the fourth step
instructionism all the way i mean our
interactionism all the way it's like
it gives you the rule walk forward so
you can replace
on a structured ascension
you can replace the sequential
instructions
with a rule or a pattern so that it's
interactionism all the way down all the
way up
and that kind of relates to active as a
scale-free framework
like bottom up from where blue
are you going to start with subatomic or
where is the kernel
going to be and maybe just
interactions all the way down and then
at some
sufficient level you can granularize or
you can coarse grain you can make
something look as if it has
instructions so i wonder if it's like
you know you see this loop right the
active inference loop and it's like an

oval
like this and so i wonder if it's like
starts to be like the structure of an
atom
like okay there's like this oval that's
this way and then i wonder if i could
superimpose one
going the other way a vertical and a
horizontal
like a little one of these things right
you know so that you know because
there's this
across scales as we scale up and down
you know
is there a loop going between and then a
loop going
this way possibly probably i don't know
i think about it in that way
so this could be the s orbital this is
just like a spherical one and maybe
there's
other trajectories or flows
that have different shapes
so dave wrote in there and anyone else
in this last
little bit is welcome to write a comment
dave wrote in
most effective military units troops are
told what to accomplish
not what to do so that's the difference
between
for me like push and pull push you know
lift the bar is like it's a command
about
what to do versus like when
it's an accomplishment and so it's the
teleos or the end
directedness that pulls the person

and then in that poll there is the space
to have the solenoidal flow to
to walk straight towards the goal
where there's a clear path but then to
be like oh yeah but then there's i'm
hitting a wall
so in order to accomplish i do have to
walk around
of course that makes sense in the
context of being pulled by a goal
but that's where the instruction-based
robotics
always fails because anytime there's
something that requires a deviation from
value
you need a total second layer of the
model to
re-introduce curiosity or epistemic gain
into the value framework so then
it's this question how do you balance
all these second layer bolt-ons
so that you can still salvage your
value-driven model
even when there's aspects that are just
clearly
better framed within a pragmatic
epistemic
distinction blue yeah it's like solving
the problem without
ground truth data right like so how do
we you know if we don't know
if we've never been exposed to the
system before if it's not programmed
into our robotic function you know how
do we um
account for that right so i think that
that's where the exploration and the
play

a lot comes in or maybe tossing and
turning
like you want to be comfortable and then
it just sort of uncoordinated
behavior that's just rummaging around
searching for something in a box or all
these examples where
it's so you couldn't say that that twist
of your side
it was it's not skillful at least for me
but it's also not reward driven again
you need a way
second level bolt-on to be explaining
how the rummaging behavior
is itself value-driven you can say that
there is a value that's pulling
or you want the valuable thing in that
box
but that's we all agree on that we all
agree that there's something valuable in
the box and that search behavior is
going to be required
are we going to have a process theory
that allows for exploration
and then zeroing in on a goal or
will we have a process theory that tries
to go like well if the end
is valuable then let's just back
propagate value
yeah or we don't know if we don't know
what's valuable in the box
right something is valuable in here like
how do we find out what is it that's
valuable okay so we implement the search
function
but then exploring what is the valuable
item out of you know 50 items in the box
right

yep it's almost like in the simplest
cases
uh the value learning is defensible
when we have this hill-climbing robot on
a single hill
okay but now you got a triple bottom
line you got the financial success
but you also have environmental and
social considerations of a business
so will you reduce all three of them
to one of them or try to generate a
fourth
standard that you're going to reduce
them all like a common currency for all
of your values and rewards
or could there be a deep generative
model where
the organization has preferences in the
financial social and environmental realm
and then take stock of its action
possibilities not even described in the
value mindset
maybe every time you have a meeting it's
like square zero what can we do
whereas in active inference we start
with the space of
policies the affordances that the niche
provides
and so you don't need to consider
policies that cannot be done
and you should re-weight policies
accordingly to
how likely or tractable they are so
there's just
so many ways that we see that the
mathematical framings
get expanded on through uh relaxation
of some of their attributes just like

first in 2011
wrote which is that optimal control can
be cached as active inference
with three simplifications
and then he walked through those which
we discussed a little bit last time but
this is kind of
cool how we're building out some
differences between the motor control
approaches that have been used in the
past and
active inference as a process theory and
then the way that
that encompasses what we've seen before
and then maybe also opens up the
discussion
into just areas that weren't even
considered like we can add in
affordances
plus preferences
that i think that embodied skillful and
an active inference implies
a huge leap of faith back to that uh
jumping off the because it is i mean
that's not
that's a that's a cliche that people can
probably appreciate
what do you think the trust or faith is
in
yeah that's it yourself
yourself right or through it
act or act not there is no infer
no there is there is um
so it's interesting yeah how how can we
distill the similarities and differences
so that when we see one of those
limitations
being like okay we had um uh curiosity

slash pragmatic plus epistemic
so that when we see a sort of here's the
new hack
on value we can see that
as a saddle point for where
that model could move towards an active
inference
framing just by kind of rewiring
some of the modules that it has
um here's a
great question from the live chat bill
wrote
would a priority parameter according to
depletion levels of different resources
be
applicable
yeah that's back to the pools thing yeah
i would say yes
okay dean with the pools here would be
my uh thought from the multi-scale
decision making let's just think about a
bee colony
the resources that they need to forage
for include
like pollen which has a lot of protein
and fat
and it's helpful for the developing
larva then nectar
which is like high octane gasoline a lot
of carbs good for the adults
um and then depending on the environment
they might need like water or salt or
other
different resources so the depletion
of uh there might be an imbalance of the
nutrients
and you can think about that similarly
for an organism like

rabbit starvation when you only eat
protein
your body will enter starvation mode
because you know other
fats and stuff are needed so definitely
multiple resources
are being balanced and part of the
challenge of decision making under
uncertainty
just as we've been getting at is that
there's not just one kind of outcome
like it's enough to be uncertain about
value but now when we're
thinking about these multi-dimensional
resource landscapes
whether it's just protein versus carbs
or whether it's a lot more nuanced like
the triple bottom line
we really do need a principled way
of talking about the relative depletion
of
different uh resources so
in the bee colony case selection
has shaped nest mates so that
they use the rate and the type of
interactions they have
in order to update their nest mate level
behavior so i'm having a lot of
interactions with hungry larva
maybe we need more pollen i'm having a
lot of interactions with
food reservoirs maybe that can slow me
down because we probably have enough
forage for now so colonies that don't
consist
of nest mates with that productive way
of interpreting their interactions
not instructions those colonies are

swept off the table
and so we end up seeing distributed
systems succeed
where they act as if there is a correct
prioritization
amidst uncertainty of different kinds of
resources
and then if we use active inference
instrumentally
in the deep generative model of the
body's
nutrient ratios or the colonies relative
balance between protein and
carbs then it's as if
there is a priority parameter where
when something is depleted it's the one
that's prioritized
but we don't need that expressively in
the model
we can have a multi-dimensional um
generative model of different resources
and then actions will be selected
according to their expected free energy
being minimized
almost going along the curve that takes
us on that edge of
explore and exploit from wherever we are
in that resource depletion landscape
towards our preferences
so would be totally open to hearing
other explanations or maybe there is a
priority parameter that can come into
play
but one sort of longitud response
would be organisms do act adaptively to
prioritize depleted resources
but that doesn't mean that there needs
to be a parameter in our model

that explicitly prioritizes specific
resources it's enough just to have a
preference vector
that includes multiple types of
resources blue
no i think that that's right i think the
um
right so you can make it as simple or
complicated
as you want but i always think less
parameters in the model is better
it just makes me think you know how do
we get those preferences
and then what happens when the niche
changes
so that preferences that were adaptive
like to
act as if prioritization of sugar and
salt over all other
preferences for example what happens
when
the situation changes so how do we
integrate high precision priors
with highly accurate sensory input
that's something we've talked about
before how do we get the best of both
worlds
with being able to like fly blind
because we have a deep generative model
but then also
when there is sensory input that
requires a categorical shift
in cognitive model for example we want
to
have the fluidity to do that and we
don't want to
be wasting our time in one mode
when it's the wrong mode we don't want

to
be hard when hardness is a flaw or be
soft when
softness is a flaw so i wonder if
there's like
um you know i mean for organisms like
bees and birds like there's some
programmed i mean things are shifting
all the time right like there's
you know a huge amount of like migratory
stuff that comes into play and
hibernating and
all these things right um and so like
for us like we
noticed that prioritized like
shift like i need water right now
like you know if we're suddenly like
parched right like the need for water is
overwhelming
but there's not this like because we
have offloaded so much of our
needs into our niche like the need for
to stay warm or whatever we build houses
we wear clothes
so we've offloaded a lot of this stuff
so i i wonder if it's like um
the instinct becomes less instinctual
i wonder if that's if that happens like
the instinct for to
sequester food or fly south for the
winter or these types of things
um just because you know we don't um
we don't we've just offloaded our need
to do those things so it just becomes
i don't know there
i think that's another uh distinction
here
here we can include extended slash

embedded you know four e's seventies
eleven e's however many there are
um we can at least frame those
situations in terms of niche
modification or niche construction where
is that
in motor control it's nowhere where is
the
golf club with a squishy handle that is
being
you know interacted with as an
instrument
we don't have those nice interfaces
whereas active inference has excellent
interfaces
quantitatively and qualitatively for
being enriched
by pieces that are as dean would say
maybe non-negotiable
for the actual creature but it'd be
totally ad-hoc where would you put
the instrumentation in here is it a
state estimator should i be estimating
where my golf club is
or is it enough just to estimate my
hands and the angle of the hands and
then do another
second level prediction not quite clear
you could probably
implement it multiple ways and here's
another nice
comment from dave so dave said the term
in cybernetics
so it's always good to connect back to
related areas
or putting drives into a model is
algodonic
so here's the definition of algodonic

comes from
i guess algos meaning pain and hedon
meaning pleasure
sorry dave for the mispronunciation um
putting multiple algodonic drives like
pains and pleasures may yield
complex instructive behaviors but there
are so many advantages in having a model
i think active inference so
sophisticated that it generates its
drives
the drives come out of the model rather
than being put in
so yes and he wrote then clarifying i
mean the innumerable
drives emerge from the model's internal
dynamics
if your robot is spontaneously curious
spontaneously attends to events or
responsive externals
it's as if it has a curiosity parameter
so it's true maybe we could even write
like drives
quote you know as if
i'm gonna i'm gonna take shakespeare
thanks hamlet you are great
a great prince um drives
drives emerge from model
um don't need to be put in so
instead of uh like maybe the rapid being
put in the hat
this is like the rabbits emerge from the
hat
if you you know put in enough hydrogen
and enough time
so that's interesting to think about
where cybernetic models
come into play here so let's give just

a few final minutes all right any final
questions
in the chat and thanks of course to
dave and to phil for the great questions
we can just think about um maybe dot one
dean is um being on the edge
dot zero is climbing
and so it's a two directional trail you
know there's people going up to half
dome and there's people coming down
so dot zero is two directional
highway lots of on-ramps and off-ramps
but we
try to make the topics accessible for
those who are
not in the active inference game but
they're related they're familiar with
related topics like the keywords usually
and in the dot zero we're going to go
from bridging
the keywords and the topics of broader
concern to active inference
so that's the on-ramp to active
inference but then also for those who
are
already in the active inference mode of
learning by doing and working then
it's their window out to like oh wow you
know this idea that i had been working
on
active inference in one context it's
also applying outwards
so that's in the dot zero dot one you
know
who whoever and whenever it is
uh we are in the the bathtub face the
part between the book ends
that's we're at the edge and we can

look back to the dot zero and see
again recapping the the steps that we
took
but also we want to start looking
out and then i don't know where dot two
would be but dot two is when we sort of
take the next step and
the dot two is the no going back i mean
i
i can add a twist to the fact that i'm
i'm going to be hitting the water soon
and that's a skillful embodied movement
right we actually watch people that
do the 10 meter platform and can add up
quite a few things before they
hit the water and regain control but i
i i i think this is this
is what interactionism is versus
instructionalism
i mean the way that this is set up the
zero one two
is interactionism it's more about this
than it is about um bi-directionality
and i think you talked about that at the
beginning of today's
episode right we have a
a structure that we hold to a rail
that we could modify as well but we hold
to the rail
so that we don't have to frame it like
instructions
um we weren't instructed to do this and
i don't know
i hope that people don't feel like these
are instructions but
rather interactions or engagements with
them and then just like so many other
things that we're seeing online

the new technological affordances allow
it to actually be an interaction we
actually
can have people asking questions live
being a part of the conversation whereas
if it were just recording a video and
releasing it or record
making a book and releasing it it'd be
hard to break out of the instructionist
frame
because the deliverable would be static
and very unidirectional here
we have a different affordance that
helps us break out of that frame
basically for free
because even if instructions are being
conveyed on the live stream
it would still be within an interaction
of participants of individual and
collective
and of the community so
any final thoughts
on one no fun times thanks both of you
for the
the dot zero and dot one fun we've had
a few hours to think about it but you
know the paper
is the distillation of many more hours
so there's always
value dare i say and
going back and playing with it because
it's fun
and it's just yep there are great topics
to be considering about and learning
so cool thanks everyone for watching
thanks dean and blue
and we'll see you next week for 23.2
thanks guys bye